CIE AS PHYSICS (9702) • QUESTION PAPER

11.2 Fundamental Particles

Syllabus Section: 11. Particle Physics **Paper Type:** Multiple Choice (Paper 1) **Total:** 22 Questions (22 Marks)

Question 1 Easy [1 mark]

What is a particle that is **not** made of any smaller particles?

A elementary particle

B minimum particle

C original particle

D atomic particle

Question 2 Easy [1 mark]

Light is made up of which type of elementary particle?

A quarks

B leptons

C photons

D bosons

Question 3 Easy [1 mark]

What type of elementary particles are electrons ?

A quarks

B leptons

C photons

D bosons

Question 4 Easy [1 mark]

What elementary particles are the building blocks for protons and neutrons ?

A quarks

B leptons

C photons

D bosons

Question 5 Easy [1 mark]

How many quarks make up a proton ?

A 0

B 1

C 2

D 3

Question 6 Easy [1 mark]

Which particle is made up of two 'down' quarks and one 'up' quark?

A atom

B electron

C neutron

D proton

Question 7 Easy [1 mark]

How many types of quark are there ?

A 2

B 4

C 6

D 8

Question 8 Easy [1 mark]

Which elementary particle is a lepton ?

A neutron

B electron

C quark

D proton

Question 9 Easy [1 mark]

What are the fundamental particles of an atom ?

A quarks, gluons and electrons

B protons, neutrons and electrons

C quarks, photons and electrons

D quarks, gluons and neutrino

Question 10 Easy [1 mark]

What are mesons ?

A a type of composite particle produced by high energy

B a type of gluon

C an antimatter version of the electron

D a type of neutrino

Question 11 Medium [1 mark]

Which of the following sentences about antimatter is **incorrect** ?

A antimatter is normal matter with an opposite charge

B antimatter is only produced in particle accelerators

C antimatter destroys matter

D equal amounts of antimatter and matter were created during the Big Bang

Question 12 Medium [1 mark]

Fundamental or elementary particles are particles that aren't made up of smaller particles.

What is the most common type of fundamental particle in the universe ?

A atom

B meson

C neutrino

D quark

Question 13 Medium [1 mark]

In a nucleus of Tritium, ${}^{3}_{1}H$, how many 'down' quarks are there?

A 5

B 4

C 3

D 2

Question 14 Medium [1 mark]

Which row in the table gives the correct quark combination for the particle?

	Particle	Category	Quark combination
A	neutron	baryon	ud
B	neutron	meson	udd
C	proton	baryon	uud
D	positive pion	meson	ud

A

B

C

D

Question 15 Medium [1 mark]

What are the numbers of hadrons, baryons and mesons in an atom of $^{7}_{3}Li$?

	Hadrons	Baryons	Mesons
A	7	3	3
B	7	4	4
C	7	7	0
D	10	7	0

A

B

C

D

Question 16 Hard [1 mark]

Quarks are thought to make up protons and neutrons.

The 'up' quark has a charge of $\frac{2}{3}e$; a 'down' quark has a charge of $-\frac{1}{3}e$, where e is the elementary charge (+ 1.6 × 10^{-19} C)

How many up quarks and down quarks must a proton contain ?

	Up quarks	Down quarks
A	0	3
B	2	1
C	1	2
D	1	1

A

B

C

D

Question 17 Hard [1 mark]

A CERN detector shows a signal only in the hadronic calorimeter. No signal is observed in the tracker, electromagnetic calorimeter or muon chambers.

Which particle must be responsible for this signal?

A pion

B neutrino

C photon

D neutron

Question 18 Hard [1 mark]

Some particles are made up of a combination of three quarks.

Which particle would **not** give a particle a charge of either $+1.6 \times 10^{-19}$ C or zero?

A up, up, up

B up, up, down

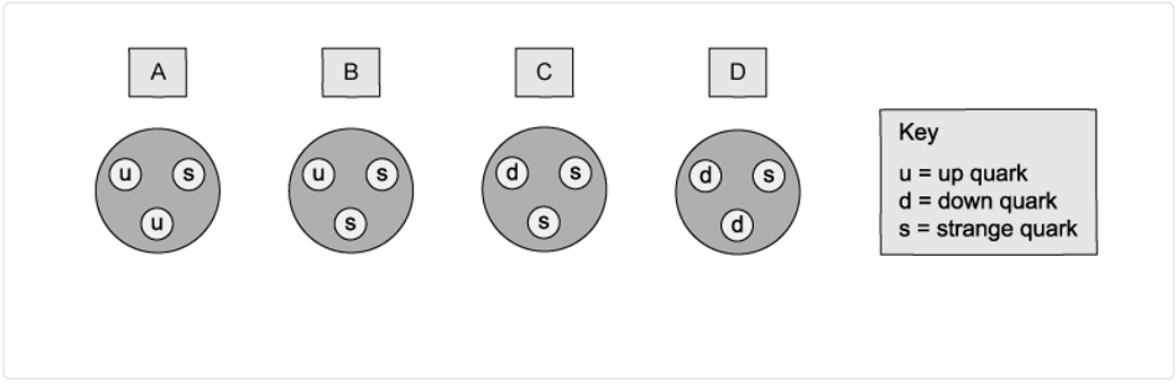
C up, strange, strange

D up, down, down

Question 19 Hard [1 mark]

The composition of four different hadrons are shown in the diagram below. One of the hadrons is a Σ^+ particle. It has a charge of $+e$, where is e is the elementary charge.

Which hadron could be the Σ^+ particle ?



A

B

C

D

Question 20 Hard [1 mark]

When high-speed hadrons collide in particle accelerators, a positive pion (π^+) is an unstable particle produced. The π^+ particle has a charge of $+e$.

What is the quark combination of the π^+ particle?

A $u \quad u$

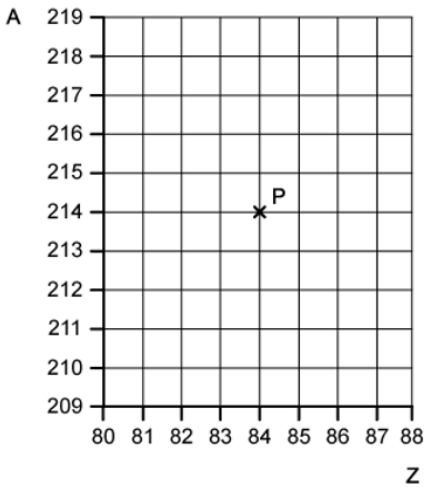
B $u \quad d$

C $d \quad u$

D $d \quad d$

Question 21 Hard [1 mark]

A graph of nucleon number A against proton number Z is shown in the diagram below.



The graph shows a cross that represents a nucleus P.

Nucleus P decays by emitting an α-particle to form a nucleus Q.

Nucleus Q then decays by emitting a β^- particle to form a nucleus R.

The quark composition of one nucleon in Q is changed during the emission of the β^- particle.

What is the corresponding change in quark composition from nucleus Q to nucleus R ?

A no change

B *u*-quark to *d*-quark

C *d*-quark to *u*-quark

D *d*-quark to *s*-quark

Question 22 Hard [1 mark]

The equation for β^- decay is

$$n \rightarrow p + e^- + v_e$$

Using your knowledge of antiparticles and the weak interaction, what is the equation for β^+ decay?

A $n \rightarrow p + e^+ + v_e$

B $p \rightarrow n + e^+ + v_e$

C $n + e^+ \rightarrow p + v_e$

D $p \rightarrow n + e^+ + v_e$